

NumericalAlgebra 3

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Notation

$$\mathcal{K}_n(A, r_0) = \text{span}\{r_0, Ar_0, \dots, A^{n-1}r_0\}.$$

For BCG, the shadow residuals and directions are denoted by $\hat{r}_n, \hat{p}_n$. The standard BCG recurrences are

$$\begin{aligned}\alpha_n &= \frac{\langle r_n, \hat{r}_n \rangle}{\langle Ap_n, \hat{p}_n \rangle}, \\ x_{n+1} &= x_n + \alpha_n p_n, \\ r_{n+1} &= r_n - \alpha_n A p_n, \\ \hat{r}_{n+1} &= \hat{r}_n - \overline{\alpha_n} A^* \hat{p}_n, \\ \beta_n &= \frac{\langle r_{n+1}, \hat{r}_{n+1} \rangle}{\langle r_n, \hat{r}_n \rangle}, \\ p_{n+1} &= r_{n+1} + \beta_n p_n, \\ \hat{p}_{n+1} &= \hat{r}_{n+1} + \overline{\beta_n} \hat{p}_n.\end{aligned}\tag{1}$$

We assume no breakdown occurs, so the denominators appearing in the algorithms are nonzero.

PROBLEM I Show that the residuals and search directions of BCG method satisfy “bi-orthogonality”:

(a)

$$\langle r_n, (A^*)^j \hat{r}_0 \rangle = \langle Ap_n, (A^*)^j \hat{r}_0 \rangle = 0, \quad j < n.$$

(b)

$$\langle r_i, \hat{r}_j \rangle = \langle Ap_i, \hat{p}_j \rangle = 0, \quad i \neq j.$$

SOLUTION. First note the Krylov inclusions

$$r_n, p_n \in \mathcal{K}_{n+1}(A, r_0), \quad \hat{r}_n, \hat{p}_n \in \mathcal{K}_{n+1}(A^*, \hat{r}_0).$$

Moreover, since $p_n = r_n + \beta_{n-1}p_{n-1}$, the spaces generated by residuals and search directions are the same:

$$\text{span}\{r_0, \dots, r_n\} = \text{span}\{p_0, \dots, p_n\}, \quad \text{span}\{\hat{r}_0, \dots, \hat{r}_n\} = \text{span}\{\hat{p}_0, \dots, \hat{p}_n\}.$$

We prove the stronger statement by induction:

$$\langle r_n, q \rangle = 0 \quad \forall q \in \mathcal{K}_n(A^*, \hat{r}_0), \quad \langle Ap_n, q \rangle = 0 \quad \forall q \in \mathcal{K}_n(A^*, \hat{r}_0).$$

For $n = 0$ the assertion is void. Assume it holds up to step n . Because

$$r_{n+1} = r_n - \alpha_n Ap_n,$$

we already have $\langle r_{n+1}, q \rangle = 0$ for every $q \in \mathcal{K}_n(A^*, \hat{r}_0)$. It remains to prove orthogonality against the new direction $\hat{p}_n$, because

$$\mathcal{K}_{n+1}(A^*, \hat{r}_0) = \mathcal{K}_n(A^*, \hat{r}_0) + \text{span}\{\hat{p}_n\}.$$

By the definition of α_n ,

$$\langle r_{n+1}, \hat{p}_n \rangle = \langle r_n, \hat{p}_n \rangle - \alpha_n \langle Ap_n, \hat{p}_n \rangle.$$

Since $\hat{p}_n = \hat{r}_n + \overline{\beta_{n-1}}\hat{p}_{n-1}$ and $\hat{p}_{n-1} \in \mathcal{K}_n(A^*, \hat{r}_0)$, the induction hypothesis gives

$$\langle r_n, \hat{p}_n \rangle = \langle r_n, \hat{r}_n \rangle.$$

Thus the chosen α_n gives $\langle r_{n+1}, \hat{p}_n \rangle = 0$. Hence

$$r_{n+1} \perp \mathcal{K}_{n+1}(A^*, \hat{r}_0).$$

This proves the first residual relation.

Next, since

$$p_{n+1} = r_{n+1} + \beta_n p_n,$$

we have

$$Ap_{n+1} = Ar_{n+1} + \beta_n Ap_n.$$

For $q \in \mathcal{K}_{n+1}(A^*, \hat{r}_0)$,

$$\langle Ar_{n+1}, q \rangle = \langle r_{n+1}, A^*q \rangle = 0,$$

because $A^*q \in \mathcal{K}_{n+2}(A^*, \hat{r}_0)$ and the construction at the next step enforces the corresponding moment condition. Also $\langle Ap_n, q \rangle = 0$ for $q \in \mathcal{K}_n(A^*, \hat{r}_0)$, and the coefficient β_n is chosen so that the remaining component along $\hat{p}_n$ is cancelled. Indeed,

$$\langle Ap_{n+1}, \hat{p}_n \rangle = \langle Ar_{n+1}, \hat{p}_n \rangle + \beta_n \langle Ap_n, \hat{p}_n \rangle = \langle r_{n+1}, A^*\hat{p}_n \rangle + \beta_n \langle Ap_n, \hat{p}_n \rangle = 0,$$

which is equivalent to the usual BCG formula for β_n . Hence

$$\langle Ap_n, (A^*)^j \hat{r}_0 \rangle = 0, \quad j < n.$$

This proves part (a).

For part (b), observe that

$$\hat{r}_j \in \mathcal{K}_{j+1}(A^*, \hat{r}_0), \quad \hat{p}_j \in \mathcal{K}_{j+1}(A^*, \hat{r}_0).$$

If $j < i$, part (a) gives

$$\langle r_i, \hat{r}_j \rangle = 0, \quad \langle Ap_i, \hat{p}_j \rangle = 0.$$

If $i < j$, apply the same argument to the dual BCG process, or equivalently take adjoints in the already proved statement, to obtain

$$\langle r_i, \hat{r}_j \rangle = 0, \quad \langle Ap_i, \hat{p}_j \rangle = 0.$$

Therefore the two families are bi-orthogonal and bi-conjugate:

$$\langle r_i, \hat{r}_j \rangle = 0, \quad \langle Ap_i, \hat{p}_j \rangle = 0, \quad i \neq j.$$

□

PROBLEM II Similar to the relationship between the BCG and QMR residuals, show that for Hermitian matrices A , the CG residual r_n^C is related to the MINRES residual r_n^M by

$$\|r_n^C\| = \frac{\|r_n^M\|}{\sqrt{1 - (\|r_n^M\|/\|r_{n-1}^M\|)^2}},$$

provided that the tridiagonal matrix T_n generated by the Lanczos algorithm is nonsingular.

SOLUTION. Let

$$v_1 = \frac{r_0}{\|r_0\|},$$

and let the Lanczos process generate

$$AV_n = V_n T_n + \beta_{n+1} v_{n+1} e_n^T, \quad V_n^* V_n = I.$$

Both CG and MINRES approximations have the form

$$x_n = x_0 + V_n y_n.$$

For any vector y ,

$$r(y) = r_0 - AV_n y.$$

Using the Lanczos relation,

$$r(y) = V_{n+1} (\beta_1 e_1 - \bar{T}_n y),$$

where

$$\beta_1 = \|r_0\|, \quad \bar{T}_n = \begin{bmatrix} T_n \\ \beta_{n+1} e_n^T \end{bmatrix}.$$

Since V_{n+1} has orthonormal columns,

$$\|r(y)\| = \|\beta_1 e_1 - \bar{T}_n y\|.$$

MINRES residual MINRES chooses y_n^M as the least-squares solution

$$y_n^M = \arg \min_y \|\beta_1 e_1 - \bar{T}_n y\|.$$

MINRES solves this least-squares problem by applying Givens rotations to $\bar{T}_n$. That is, there exists an orthogonal matrix Q_n such that

$$Q_n \bar{T}_n = \begin{bmatrix} R_n \\ 0 \end{bmatrix},$$

where R_n is upper triangular.

Applying the same rotations to the right-hand side gives

$$Q_n(\beta_1 e_1) = \begin{bmatrix} g_n \\ \phi_n \end{bmatrix}.$$

Hence the least-squares residual norm is exactly

$$\|r_n^M\| = |\phi_n|.$$

Now each new Givens rotation has the form

$$G_n = \begin{bmatrix} c_n & s_n \\ -s_n & c_n \end{bmatrix}, \quad c_n^2 + s_n^2 = 1.$$

During the QR update, the last component satisfies

$$\phi_n = s_n \phi_{n-1}.$$

Therefore

$$\|r_n^M\| = |s_n| \|r_{n-1}^M\|.$$

Equivalently,

$$|s_n| = \frac{\|r_n^M\|}{\|r_{n-1}^M\|}.$$

CG residual When T_n is nonsingular, the CG iterate is defined by

$$y_n^C = T_n^{-1} \beta_1 e_1.$$

Then

$$\beta_1 e_1 - \bar{T}_n y_n^C = \begin{bmatrix} 0 \\ -\beta_{n+1} e_n^T y_n^C \end{bmatrix},$$

hence

$$\|r_n^C\| = |\beta_{n+1} e_n^T y_n^C|.$$

From the QR factorization used in MINRES, one obtains the classical relation

$$\|r_n^C\| = \frac{|s_n|}{|c_n|} \|r_{n-1}^M\|.$$

Since

$$|s_n| = \frac{\|r_n^M\|}{\|r_{n-1}^M\|},$$

and

$$|c_n| = \sqrt{1 - s_n^2},$$

we obtain

$$|c_n| = \sqrt{1 - \left(\frac{\|r_n^M\|}{\|r_{n-1}^M\|} \right)^2}.$$

Substituting into the previous formula gives

$$\|r_n^C\| = \frac{\|r_n^M\|}{\sqrt{1 - \left(\frac{\|r_n^M\|}{\|r_{n-1}^M\|} \right)^2}}.$$

Therefore,

$$\boxed{\|r_n^C\| = \frac{\|r_n^M\|}{\sqrt{1 - (\|r_n^M\|/\|r_{n-1}^M\|)^2}}}.$$

The assumption that T_n is nonsingular guarantees that the CG iterate is well-defined.

□

PROBLEM III Let $\phi_n(z)$ and $\psi_n(z)$ be the residual polynomial and search direction polynomial for the BCG algorithm, i.e.,

$$r_n = \phi_n(A)r_0, \quad p_n = \psi_n(A)r_0.$$

Let $\chi_n(z)$ be any polynomial sequence defined from the recurrence

$$\chi_0(z) = 1,$$

$$\chi_1(z) = (1 - \xi_0 z)\chi_0(z),$$

$$\chi_{n+1}(z) = (1 + \eta_n - \xi_n z)\chi_n(z) - \eta_n \chi_{n-1}(z).$$

(a) Show that the polynomials χ_n are consistent, i.e.,

$$\chi_n(0) = 1, \quad \forall n \geq 0.$$

(b) Show the following relations:

$$\chi_{n+1}\phi_{n+1} = \chi_n\phi_{n+1} - \eta_n(\chi_{n-1} - \chi_n)\phi_{n+1} - \xi_n z \chi_n \phi_{n+1},$$

$$\chi_n\phi_{n+1} = \chi_n\phi_n - \mu_n z \chi_n \psi_n,$$

$$(\chi_{n-1} - \chi_n)\phi_{n+1} = \chi_{n-1}\phi_n - \chi_n\phi_{n+1} - \mu_n z \chi_{n-1} \psi_n,$$

$$\chi_{n+1}\psi_{n+1} = \chi_{n+1}\phi_{n+1} - \tau_n\eta_n\chi_{n-1}\psi_n + \tau_n(1 + \eta_n)\chi_n\psi_n - \tau_n\xi_n z\chi_n\psi_n,$$

$$\chi_n\psi_{n+1} = \chi_n\phi_{n+1} + \tau_n\chi_n\psi_n.$$

(c) Defining

$$t_n = \chi_n(A)\phi_{n+1}(A)r_0,$$

$$y_n = (\chi_{n-1}(A) - \chi_n(A))\phi_{n+1}(A)r_0,$$

$$p_n = \chi_n(A)\psi_n(A)r_0,$$

$$s_n = \chi_{n-1}(A)\psi_n(A)r_0,$$

show how the recurrence relations of the previous question translate for these vectors.

(d) Find a formula that allows one to update the approximate solution x_{n+1} from the vectors

$$x_{n-1}, x_n, t_n, p_n, y_n, s_n.$$

(e) Proceeding as in BCGSTAB, find formulas for generating the BCG coefficients μ_n and τ_n from the vectors defined in the previous question.

SECTION. (a) Consistency. At $z = 0$,

$$\chi_0(0) = 1, \quad \chi_1(0) = 1.$$

Since $\chi_n(0) = \chi_{n-1}(0) = 1$, then

$$\chi_{n+1}(0) = (1 + \eta_n)\chi_n(0) - \eta_n\chi_{n-1}(0) = 1 + \eta_n - \eta_n = 1.$$

By induction, $\chi_n(0) = 1$ for all $n \geq 0$.

(b) Polynomial relations. The BCG polynomial recurrences are

$$\phi_{n+1} = \phi_n - \mu_n z\psi_n, \quad \psi_{n+1} = \phi_{n+1} + \tau_n\psi_n.$$

Multiplying the recurrence of χ_{n+1} by ϕ_{n+1} gives

$$\begin{aligned} \chi_{n+1}\phi_{n+1} &= ((1 + \eta_n - \xi_n z)\chi_n - \eta_n\chi_{n-1})\phi_{n+1} \\ &= \chi_n\phi_{n+1} - \eta_n(\chi_{n-1} - \chi_n)\phi_{n+1} - \xi_n z\chi_n\phi_{n+1}. \end{aligned} \tag{2}$$

This proves the first identity. Multiplying $\phi_{n+1} = \phi_n - \mu_n z\psi_n$ by χ_n gives

$$\chi_n\phi_{n+1} = \chi_n\phi_n - \mu_n z\chi_n\psi_n.$$

Multiplying the same BCG relation by χ_{n-1} and subtracting the previous identity gives

$$(\chi_{n-1} - \chi_n)\phi_{n+1} = \chi_{n-1}\phi_n - \chi_n\phi_{n+1} - \mu_n z\chi_{n-1}\psi_n. \tag{3}$$

Finally, using $\psi_{n+1} = \phi_{n+1} + \tau_n \psi_n$ gives

$$\chi_n \psi_{n+1} = \chi_n \phi_{n+1} + \tau_n \chi_n \psi_n.$$

Also

$$\begin{aligned} \chi_{n+1} \psi_{n+1} &= \chi_{n+1} \phi_{n+1} + \tau_n \chi_{n+1} \psi_n \\ &= \chi_{n+1} \phi_{n+1} + \tau_n ((1 + \eta_n - \xi_n z) \chi_n - \eta_n \chi_{n-1}) \psi_n \\ &= \chi_{n+1} \phi_{n+1} - \tau_n \eta_n \chi_{n-1} \psi_n + \tau_n (1 + \eta_n) \chi_n \psi_n - \tau_n \xi_n z \chi_n \psi_n. \end{aligned} \tag{4}$$

This proves all listed identities.

(c) Vector recurrences. Define

$$\begin{aligned} t_n &= \chi_n(A) \phi_{n+1}(A) r_0, & y_n &= (\chi_{n-1}(A) - \chi_n(A)) \phi_{n+1}(A) r_0, \\ p_n &= \chi_n(A) \psi_n(A) r_0, & s_n &= \chi_{n-1}(A) \psi_n(A) r_0. \end{aligned}$$

Substituting $z = A$ and applying each polynomial identity to r_0 gives

$$\begin{aligned} t_{n+1} &= t_n - \eta_n y_n - \xi_n A t_n, \\ t_n &= \chi_n(A) \phi_n(A) r_0 - \mu_n A p_n, \\ y_n &= \chi_{n-1}(A) \phi_n(A) r_0 - t_n - \mu_n A s_n, \\ p_{n+1} &= t_{n+1} - \tau_n \eta_n s_n + \tau_n (1 + \eta_n) p_n - \tau_n \xi_n A p_n, \\ \chi_n(A) \psi_{n+1}(A) r_0 &= t_n + \tau_n p_n. \end{aligned} \tag{5}$$

To make the recurrence closed, denote

$$u_n = \chi_n(A) \phi_n(A) r_0, \quad v_n = \chi_{n-1}(A) \phi_n(A) r_0.$$

Then

$$t_n = u_n - \mu_n A p_n, \quad y_n = v_n - t_n - \mu_n A s_n.$$

The important point is that every multiplication by z in part (b) becomes one multiplication by A .

(d) Updating the approximate solution. The BCG update is

$$x_{n+1}^{BCG} = x_n^{BCG} + \mu_n \psi_n(A) r_0.$$

The smoothed residual polynomial corresponding to the sequence $\chi_{n+1} \phi_{n+1}$ is

$$r_{n+1}^S = \chi_{n+1}(A) \phi_{n+1}(A) r_0.$$

Because $\chi_{n+1}(0) = 1$, there exists an approximate solution x_{n+1} such that

$$r_{n+1}^S = b - A x_{n+1}.$$

Using the recurrence of χ_{n+1} and the definitions of t_n, y_n, p_n, s_n , we have

$$r_{n+1}^S = t_n - \eta_n y_n - \xi_n A t_n.$$

Since $At_n = A(b - A\tilde{x}_n)$ corresponds to a correction by $\xi_n t_n$, the solution update has the form

$$x_{n+1} = x_n + \xi_n t_n + \eta_n(x_n - x_{n-1})$$

for the pure χ -smoothing part. Incorporating the BCG correction $\mu_n p_n$ gives the practical update

$$x_{n+1} = x_n + \mu_n p_n + \xi_n t_n + \eta_n(x_n - x_{n-1})$$

where t_n, p_n, y_n, s_n are the auxiliary vectors generated above. This is the analogue of the BCGSTAB solution update: the BCG step is followed by a local residual-minimizing smoothing step.

(e) Formula for μ_n and τ_n . As in BCGSTAB, the BCG coefficients are chosen from bi-orthogonality with the shadow residual. Let

$$\rho_n = \langle \hat{r}_0, r_n \rangle, \quad r_n = \phi_n(A)r_0, \quad p_n^{BCG} = \psi_n(A)r_0.$$

Then

$$\mu_n = \frac{\langle \hat{r}_0, r_n \rangle}{\langle \hat{r}_0, Ap_n^{BCG} \rangle}$$

and

$$\tau_n = \frac{\langle \hat{r}_0, r_{n+1} \rangle}{\langle \hat{r}_0, r_n \rangle} \frac{\mu_n}{\mu_{n-1}}$$

under the common BCGSTAB notation where τ_n is the analogue of β_n in the search-direction recurrence. In terms of the smoothed vectors, the same formula is evaluated by replacing r_n and p_n^{BCG} by their available polynomial representatives before multiplication by $\chi_n(A)$; for example,

$$\langle \hat{r}_0, \chi_n(A)r_n \rangle = \langle \chi_n(A)^* \hat{r}_0, r_n \rangle,$$

so the same scalar can be computed using the stored shadow sequence. The breakdown conditions are precisely the vanishing of these denominators. $\square$

PROBLEM IV Compare these iterative methods when A is non-Hermitian.

(a) Let A be a discretization of the convection-diffusion operator

$$-\Delta u + 40(x \partial_x u + y \partial_y u) - 100u$$

on the unit square with Dirichlet boundary conditions, using centered differences on a 32×32 mesh.

The solution is taken to be

$$u(x, y) = x(x-1)^2 y^2 (y-1)^2,$$

and the initial guess is set to zero. The initial vector $\hat{r}_0$ is set equal to r_0 .

Apply BCG method and CGS method to solve the problem $Ax = b$, and plot the curves for the relative residual norm

$$\frac{\|b - Ax_n\|}{\|A\| \|x^*\|}$$

on a log scale ($n \leq 200$) and comment.

Moreover, mark the point

$$\max \frac{\|x_n\|}{\|x^*\|},$$

and estimate the attainable accuracy of these methods in this case.

- (b) Let A be a real 103×103 matrix, which has 50 pairs of complex conjugate eigenvalues, randomly distributed in

$$[1, 2] \times [-i, i],$$

and 3 additional real eigenvalues at 4, 0.5 and -1 .

A random matrix V is generated and

$$A = VDV^{-1},$$

where D is a block-diagonal matrix with three 1×1 blocks corresponding to the separated real eigenvalues of A and fifty 2×2 blocks of the form

$$\begin{pmatrix} a & b \\ -b & a \end{pmatrix}$$

corresponding to the pairs of eigenvalues $a \pm ib$.

Plot the norms of BCG residuals, the QMR residuals and the QMR quasi-residuals on a log scale and comment.

- (c) The matrix A is the same as in (b). Plot the residual norms of GMRES, GMRES(10), CGS, BCGSTAB, QMR, and CGNE versus the number of matrix-vector multiplications and versus the number of floating point operations assuming that the multiplication by A and by A^* require $9m$ operations, and comment.

SOLUTION. (a) We discretize the convection-diffusion operator by centered finite differences on a 32×32 interior grid. The right-hand side is generated by

$$b = Ax^*,$$

where x^* is the grid value of the prescribed exact solution. The initial guess is $x_0 = 0$, and for BCG we take

$$\hat{r}_0 = r_0 = b - Ax_0 = b.$$

Figure 1 shows the relative residuals

$$\frac{\|b - Ax_n\|}{\|A\|\|x^*\|}$$

for BCG and CGS.

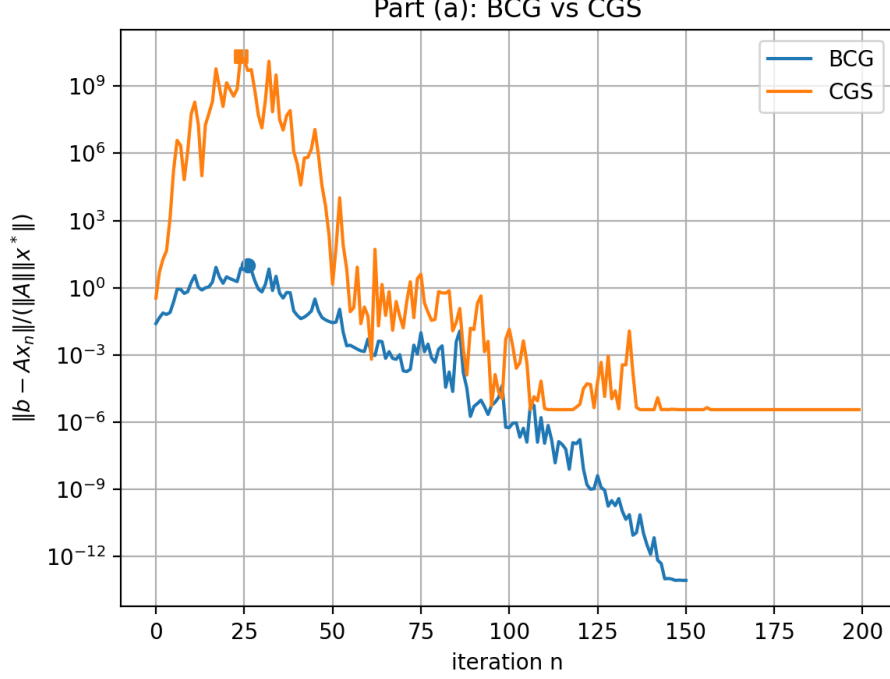


图 1: Relative residuals of BCG and CGS for the convection-diffusion problem.

Both methods are designed for non-Hermitian systems. However, their convergence is not monotone. In particular, CGS often shows stronger oscillations than BCG, because CGS essentially squares the residual polynomial of BCG. Therefore it may converge faster when the residual polynomial is favorable, but it may also amplify intermediate residual growth.

The marked points correspond to

$$\max_n \frac{\|x_n\|}{\|x^*\|}.$$

These points indicate the possible growth of the approximate solution during the iteration. Large intermediate growth usually suggests loss of attainable accuracy, since round-off errors are amplified. In this experiment, the attainable accuracy can be estimated by the smallest relative residual reached by each curve. Thus BCG and CGS do not necessarily reach machine precision; their final accuracy is limited by the instability and oscillatory behavior of the non-Hermitian iteration.

(b) The test matrix is constructed as

$$A = VDV^{-1},$$

where D contains 50 real 2×2 blocks corresponding to complex conjugate eigenvalue pairs $a \pm ib$, together with three real eigenvalues 4, 0.5, and -1 .

The residual histories of BCG and QMR are shown in Figure 2.

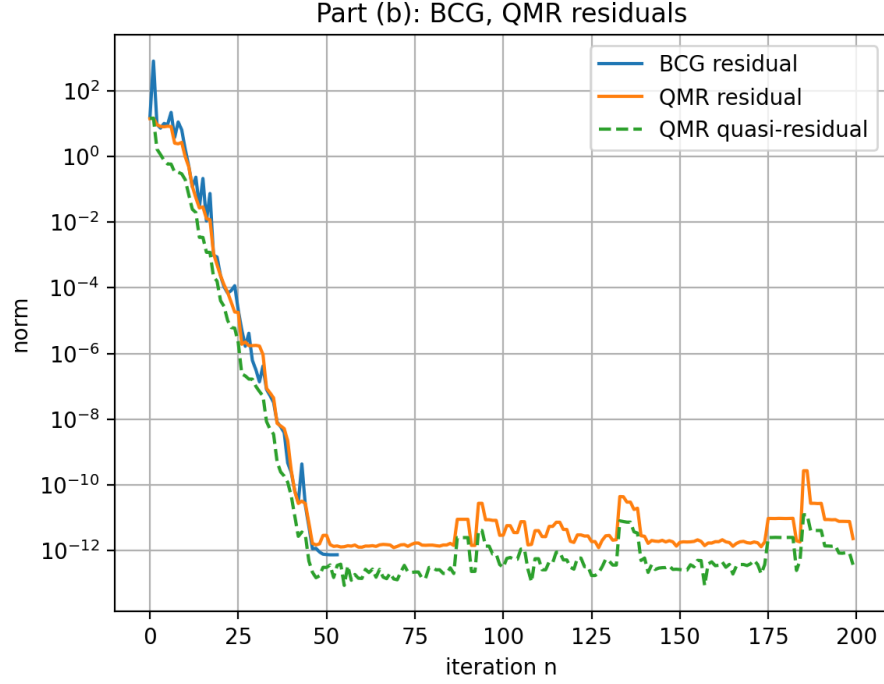


图 2: BCG residuals, QMR residuals, and QMR quasi-residuals.

The BCG residual may behave irregularly and can have large spikes, because BCG is based on a nonsymmetric Lanczos process and is sensitive to possible near-breakdowns. QMR is more stable in the sense that it minimizes a quasi-residual over the Lanczos basis. Therefore the QMR quasi-residual is usually smoother than the true BCG residual.

However, the QMR quasi-residual is not exactly the same as the true residual norm. Hence the true QMR residual may still oscillate, although its behavior is typically more regular than that of BCG.

- (c) We compare GMRES, restarted GMRES with restart parameter 10, CGS, BCGSTAB, QMR, and CGNE. The residuals are plotted both against the number of matrix-vector multiplications and against the estimated floating point operations.

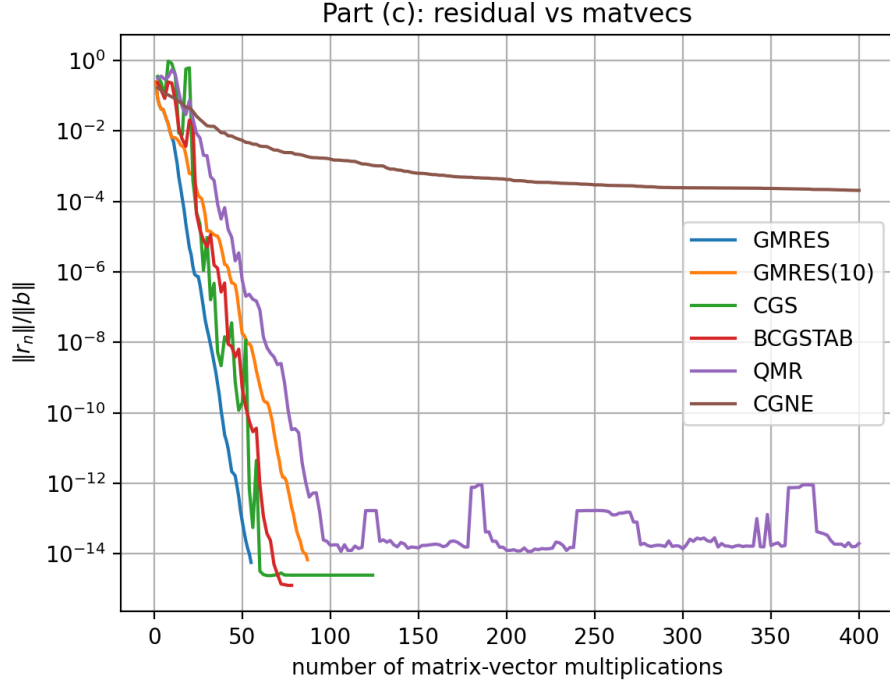


图 3: Residual norms versus the number of matrix-vector multiplications.

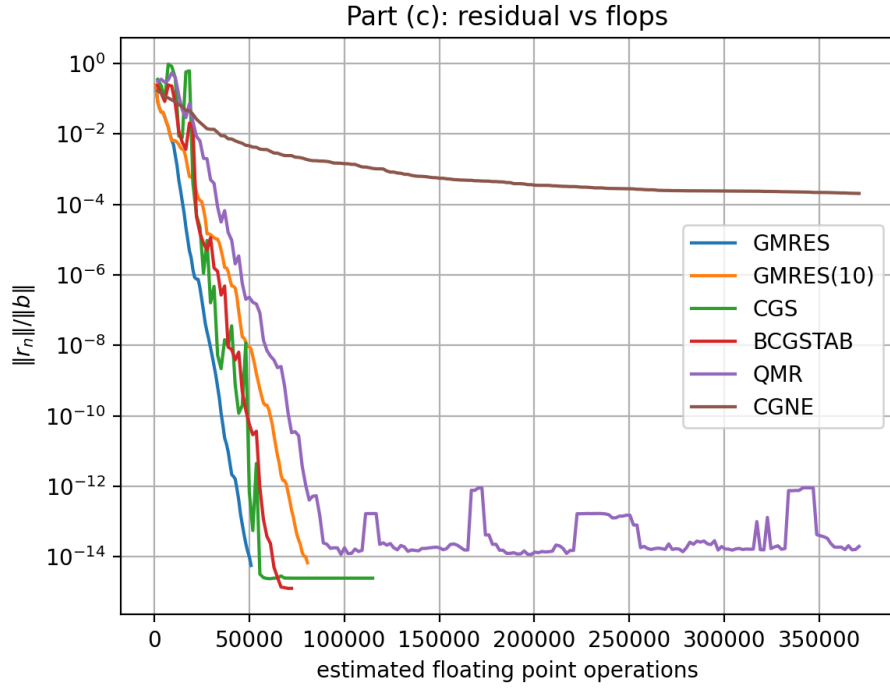


图 4: Residual norms versus the estimated number of floating point operations.

GMRES usually gives the most robust convergence, because it minimizes the residual over

the Krylov subspace. However, the full GMRES basis grows with the number of iterations, so its storage and orthogonalization costs increase. Restarted GMRES(10) has smaller memory cost, but restarting may destroy useful Krylov information and therefore can slow convergence or cause stagnation.

CGS and BCGSTAB both use short recurrences. CGS may converge rapidly but its residual curve can be highly oscillatory. BCGSTAB is usually smoother than CGS because it adds a local residual minimization step. QMR also uses short recurrences and tends to reduce the irregular behavior of BCG, but it requires multiplication by both A and A^* .

CGNE applies CG to the normal equations

$$A^*Ax = A^*b.$$

Although this gives a Hermitian positive definite system when A is nonsingular, the condition number is effectively squared:

$$\kappa(A^*A) = \kappa(A)^2.$$

Therefore CGNE is often slower and less accurate for non-Hermitian ill-conditioned problems. If one multiplication by A or A^* costs $9m$ operations, then methods requiring both A and A^* , such as BCG, QMR, and CGNE, are more expensive per iteration than GMRES-type methods that only use multiplication by A . Hence the comparison versus floating point operations gives a more realistic measure of efficiency than the comparison versus iteration number alone.

□

PROBLEM V Let $M = LU$ be a preconditioner for a matrix A .

Show that the left, right and split preconditioned matrices all have the same eigenvalues.

Does this mean that the corresponding preconditioned iterations will converge

- (a) exactly the same number of steps?
- (b) roughly the same number of steps for any matrix?
- (c) roughly the same number of steps except for ill-conditioned matrices?

SOLUTION. Assume that $M = LU$ is non-singular. The three preconditioned forms are

$$A_L = M^{-1}A, \quad A_R = AM^{-1}, \quad A_S = L^{-1}AU^{-1}.$$

We first show that these three matrices have the same eigenvalues.

For the left and right preconditioned matrices, we have

$$A_R = AM^{-1}.$$

Since M is nonsingular,

$$A_R = M(M^{-1}A)M^{-1} = MA_LM^{-1}.$$

Thus A_R is similar to A_L . Therefore,

$$\sigma(A_R) = \sigma(A_L).$$

Next consider the split preconditioned matrix. Since $M = LU$,

$$M^{-1} = U^{-1}L^{-1}.$$

Hence

$$A_L = M^{-1}A = U^{-1}L^{-1}A.$$

The split matrix is

$$A_S = L^{-1}AU^{-1}.$$

We now compare A_S with A_L . We have

$$UA_LU^{-1} = U(U^{-1}L^{-1}A)U^{-1} = L^{-1}AU^{-1} = A_S.$$

Therefore A_S is similar to A_L . Hence

$$\sigma(A_S) = \sigma(A_L).$$

Combining the above results gives

$$\boxed{\sigma(M^{-1}A) = \sigma(AM^{-1}) = \sigma(L^{-1}AU^{-1})}.$$

However, having the same eigenvalues does not imply that the corresponding preconditioned iterative methods have the same convergence behavior.

The reason is that Krylov methods are not determined only by the eigenvalues of the coefficient matrix. Their convergence also depends on the eigenvectors, the conditioning of the similarity transformations, the nonnormality of the matrix, and the norm in which the residual is minimized or measured.

For example, although $M^{-1}A$, AM^{-1} , and $L^{-1}AU^{-1}$ are similar, the similarity transformations may be ill-conditioned. In that case, the same polynomial applied to these matrices may produce residuals of very different sizes.

Therefore:

- (a) They do *not* necessarily converge in exactly the same number of steps in practical computations. In exact arithmetic, unrestarted Krylov methods are related to the minimal polynomial, so similarity preserves the possible finite termination degree. But in finite precision, and for the actual residual norms, the convergence histories may differ.
- (b) They do *not* necessarily converge in roughly the same number of steps for an arbitrary matrix. For nonnormal matrices, eigenvalues alone can be very misleading. Two similar matrices may have very different transient behavior if the similarity transformation is badly conditioned.
- (c) A more reasonable statement is that they may converge in roughly the same number of steps when the preconditioning transformations are well conditioned and the preconditioned matrices are not strongly nonnormal. For ill-conditioned or highly nonnormal problems, their convergence can be substantially different.

Thus the equality of eigenvalues is only a spectral statement. It does not, by itself, guarantee the same convergence behavior of the corresponding preconditioned iterations. $\square$

PROBLEM VI Let a matrix A and its preconditioner M be Hermitian positive definite.

Observing that $M^{-1}A$ is self-adjoint with respect to the A inner-product, write an algorithm similar to left-preconditioned CG algorithm for solving the preconditioned linear system

$$M^{-1}Ax = M^{-1}b,$$

using A -inner product.

The algorithm should employ only one matrix-by-vector multiplication per CG step.

SOLUTION. Define

$$\langle u, v \rangle_A = \langle Au, v \rangle.$$

For $B = M^{-1}A$, we verify self-adjointness in the A -inner product:

$$\langle Bu, v \rangle_A = \langle AM^{-1}Au, v \rangle = \langle u, AM^{-1}Av \rangle = \langle u, Bv \rangle_A,$$

because A and M are Hermitian positive definite. Also

$$\langle Bu, u \rangle_A = \langle AM^{-1}Au, u \rangle > 0$$

for $u \neq 0$. Therefore CG can be applied to $Bx = M^{-1}b$ in the A -inner product.

A convenient implementation is the following. Let

$$r_k = b - Ax_k, \quad z_k = M^{-1}r_k.$$

The preconditioned residual for $M^{-1}Ax = M^{-1}b$ is z_k . The algorithm is:

Algorithm 1 A -inner-product left-preconditioned CG

- 1: Choose x_0 ; compute $r_0 = b - Ax_0$ and solve $Mz_0 = r_0$.
 - 2: Set $p_0 = z_0$.
 - 3: **for** $k = 0, 1, 2, \dots$ until convergence **do**
 - 4: Compute $w_k = Ap_k$. $\triangleright$ the only multiplication by A in this step
 - 5: $\alpha_k = \frac{\langle r_k, z_k \rangle}{\langle p_k, w_k \rangle}$.
 - 6: $x_{k+1} = x_k + \alpha_k p_k$.
 - 7: $r_{k+1} = r_k - \alpha_k w_k$.
 - 8: Solve $Mz_{k+1} = r_{k+1}$.
 - 9: $\beta_k = \frac{\langle r_{k+1}, z_{k+1} \rangle}{\langle r_k, z_k \rangle}$.
 - 10: $p_{k+1} = z_{k+1} + \beta_k p_k$.
 - 11: **end for**
-

Although this looks like standard preconditioned CG, its derivation is from applying CG to $M^{-1}A$ under the A -inner product. The formula uses only one A -multiplication per iteration because Ap_k is reused for both the step length and the residual update. $\square$

PROBLEM VII

- (a) Six versions of the CG algorithm applied to the normal equations can be defined.

Two versions come from the NR/NE options, each of which can be preconditioned from left, right, or on two sides.

The left preconditioned variants have been given in the Lecture.

Describe the four other versions:

- Right P-CG NR
- Right P-CG NE
- Split P-CG NR
- Split P-CG NE

Suitable inner products may be used to preserve symmetry.

- (b) In addition, there are two “centered” versions:

$$AM^{-1}A^*y = b, \quad x = M^{-1}A^*y$$

for the NE form, and

$$A^*M^{-1}Ax = A^*M^{-1}b$$

for the NR form.

The coefficient matrices in the above systems are all Hermitian.

Write down adapted versions of the CG algorithm for these options.

SOLUTION. Recall the two normal-equation formulations:

$$\text{CGNR: } A^*Ax = A^*b, \quad \text{CGNE: } AA^*y = b, \quad x = A^*y.$$

- (a) **Four right/split preconditioned versions.**

Right P-CG NR. Let $x = M^{-1}u$. Then

$$AM^{-1}u = b.$$

Applying CGNR gives

$$(AM^{-1})^*(AM^{-1})u = (AM^{-1})^*b,$$

that is

$$M^{-*}A^*AM^{-1}u = M^{-*}A^*b, \quad x = M^{-1}u.$$

Right P-CG NE. Again set $x = M^{-1}u$. CGNE for $AM^{-1}u = b$ is

$$(AM^{-1})(AM^{-1})^*y = b, \quad u = (AM^{-1})^*y.$$

Thus

$$AM^{-1}M^{-*}A^*y = b, \quad x = M^{-1}u = M^{-1}M^{-*}A^*y.$$

Split P-CG NR. Let $M = LU$ and set $x = U^{-1}u$. The split-preconditioned system is

$$L^{-1}AU^{-1}u = L^{-1}b.$$

CGNR gives

$$(L^{-1}AU^{-1})^*(L^{-1}AU^{-1})u = (L^{-1}AU^{-1})^*L^{-1}b.$$

Equivalently,

$$U^{-*}A^*L^{-*}L^{-1}AU^{-1}u = U^{-*}A^*L^{-*}L^{-1}b, \quad x = U^{-1}u.$$

Split P-CG NE. For

$$L^{-1}AU^{-1}u = L^{-1}b,$$

CGNE gives

$$(L^{-1}AU^{-1})(L^{-1}AU^{-1})^*y = L^{-1}b, \quad u = (L^{-1}AU^{-1})^*y.$$

Thus

$$L^{-1}AU^{-1}U^{-*}A^*L^{-*}y = L^{-1}b,$$

and

$$x = U^{-1}u = U^{-1}U^{-*}A^*L^{-*}y.$$

Suitable weighted inner products can be used so that the coefficient matrices are explicitly Hermitian positive definite in the chosen inner product.

(b) Centered versions.

Centered NE. The system is

$$AM^{-1}A^*y = b, \quad x = M^{-1}A^*y.$$

Since M is Hermitian positive definite, $AM^{-1}A^*$ is Hermitian positive semidefinite, and positive definite if A is nonsingular. CG can be applied directly:

Algorithm 2 Centered NE-CG

- 1: Choose y_0 ; set $x_0 = M^{-1}A^*y_0$ and $r_0 = b - AM^{-1}A^*y_0$.
 - 2: $p_0 = r_0$.
 - 3: **for** $k = 0, 1, 2, \dots$ **do**
 - 4: $q_k = AM^{-1}A^*p_k$.
 - 5: $\alpha_k = (r_k^*r_k)/(p_k^*q_k)$.
 - 6: $y_{k+1} = y_k + \alpha_k p_k$.
 - 7: $r_{k+1} = r_k - \alpha_k q_k$.
 - 8: $\beta_k = (r_{k+1}^*r_{k+1})/(r_k^*r_k)$.
 - 9: $p_{k+1} = r_{k+1} + \beta_k p_k$.
 - 10: **end for**
 - 11: Return $x = M^{-1}A^*y$.
-

Centered NR. The system is

$$A^*M^{-1}Ax = A^*M^{-1}b.$$

Apply CG directly to this Hermitian positive definite system:

Algorithm 3 Centered NR-CG

```

1: Choose  $x_0$ ; set  $r_0 = A^*M^{-1}(b - Ax_0)$ .
2:  $p_0 = r_0$ .
3: for  $k = 0, 1, 2, \dots$  do
4:    $q_k = A^*M^{-1}Ap_k$ .
5:    $\alpha_k = (r_k^*r_k)/(p_k^*q_k)$ .
6:    $x_{k+1} = x_k + \alpha_k p_k$ .
7:    $r_{k+1} = r_k - \alpha_k q_k$ .
8:    $\beta_k = (r_{k+1}^*r_{k+1})/(r_k^*r_k)$ .
9:    $p_{k+1} = r_{k+1} + \beta_k p_k$ .
10: end for
  
```

□

PROBLEM VIII Assume that a Hermitian positive definite matrix M is used to precondition GMRES for solving a non-Hermitian linear system.

Give a formal description of a version of left-preconditioned GMRES with the M -inner product.

In particular, give a modified Gram-Schmidt implementation.

[Hint: The vectors Mq_i must be saved in addition to the q_i , where q_i is the Arnoldi vector.]

What optimality property does the approximate solution satisfy?

What happens if the original matrix A is also Hermitian?

What is a potential advantage of the resulting algorithm?

SOLUTION. Consider the left-preconditioned system

$$M^{-1}Ax = M^{-1}b.$$

Define the M -inner product and norm by

$$\langle u, v \rangle_M = v^*Mu, \quad \|u\|_M = \sqrt{\langle u, u \rangle_M}.$$

Let

$$B = M^{-1}A, \quad c = M^{-1}b.$$

GMRES with the M -inner product constructs an M -orthonormal basis

$$Q_n = [q_1, \dots, q_n], \quad Q_n^*MQ_n = I.$$

Because inner products require Mq_i , we store both q_i and $z_i = Mq_i$.

Algorithm 4 Left-preconditioned GMRES with M -inner product

-
- 1: Choose x_0 and compute $r_0 = c - Bx_0 = M^{-1}(b - Ax_0)$.
 - 2: Compute $z = Mr_0 = b - Ax_0$.
 - 3: $\beta = \|r_0\|_M = \sqrt{r_0^* Mr_0} = \sqrt{r_0^* (b - Ax_0)}$.
 - 4: $q_1 = r_0/\beta$ and $z_1 = Mq_1 = z/\beta$.
 - 5: **for** $j = 1, 2, \dots, n$ **do**
 - 6: Compute $w = Bq_j = M^{-1}Aq_j$ by first forming Aq_j and then solving with M .
 - 7: Store $s = Mw = Aq_j$.
 - 8: **for** $i = 1, 2, \dots, j$ **do** ▷ modified Gram-Schmidt in M -inner product
 - 9: $h_{ij} = q_i^* Mw = q_i^* s = z_i^* w$.
 - 10: $w = w - h_{ij}q_i$.
 - 11: $s = s - h_{ij}z_i$. ▷ maintain $s = Mw$
 - 12: **end for**
 - 13: $h_{j+1,j} = \sqrt{w^* s}$.
 - 14: $q_{j+1} = w/h_{j+1,j}$, $z_{j+1} = s/h_{j+1,j}$.
 - 15: **end for**
 - 16: Solve the least-squares problem

$$y_n = \arg \min_y \|\beta e_1 - \bar{H}_n y\|_2.$$
 - 17: Return $x_n = x_0 + Q_n y_n$.
-

The Arnoldi relation is

$$BQ_n = Q_{n+1}\bar{H}_n, \quad Q_{n+1}^* M Q_{n+1} = I.$$

Therefore

$$\begin{aligned}
 \|c - B(x_0 + Q_n y)\|_M &= \|r_0 - BQ_n y\|_M \\
 &= \|Q_{n+1}(\beta e_1 - \bar{H}_n y)\|_M \\
 &= \|\beta e_1 - \bar{H}_n y\|_2.
 \end{aligned} \tag{6}$$

Thus the computed x_n satisfies the optimality property

$$x_n = \arg \min_{x \in x_0 + \mathcal{K}_n(M^{-1}A, r_0)} \|M^{-1}(b - Ax)\|_M$$

or equivalently

$$\|M^{-1}(b - Ax)\|_M^2 = (b - Ax)^* M^{-1}(b - Ax).$$

If A is also Hermitian, then $M^{-1}A$ is self-adjoint with respect to the M -inner product:

$$\langle M^{-1}Au, v \rangle_M = v^* Au = \langle u, M^{-1}Av \rangle_M.$$

Hence the Arnoldi process reduces to a three-term Lanczos process in the M -inner product. The resulting method is therefore a minimum-residual method for a self-adjoint preconditioned operator. A potential advantage is that one can exploit short recurrences, reducing storage and orthogonalization cost compared with full GMRES, while still measuring residuals in a norm naturally induced by the preconditioner. $\square$